

Lecture 12

Kinetics of Rigid Bodies

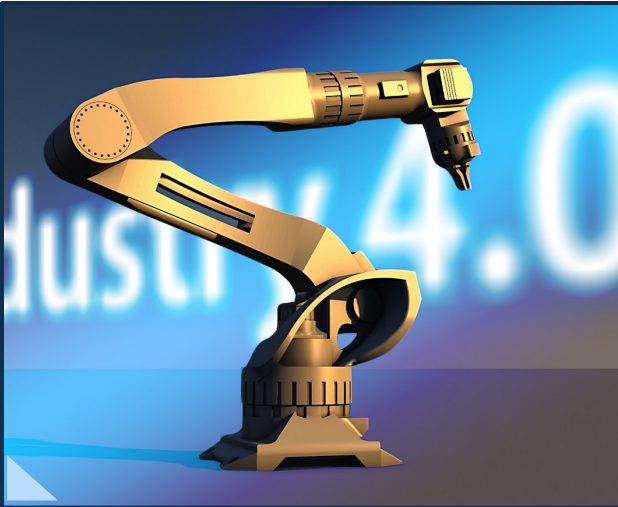
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MC2103
Fall 2026



12 Kinetics of Rigid Bodies

Introduction

In the previous lectures, we studied kinematics for rigid bodies in 3D space. We also studied kinetics for particles. In this lecture, we will study kinetics for rigid bodies in 3D space.



Equations of Motion for Rigid Bodies

⚙ Particle vs Rigid Bodies

- We learned that Newton's law of motion ($F=ma$) can be applied to each particle within a system of multiple particles.
- We also learned how to derive equations of motion for particles by drawing free body diagrams.
- However, in 3D space, drawing free body diagrams is not as simple as it was in the previous lectures.

Equations of Motion for Rigid Bodies

⚙ Particle vs Rigid Bodies

- Even though the fundamental principle is still the same, we will learn how to derive equations of motion via algebraic methods.
- In 3D space, equations of motion for translation will be mostly derived by linear momentum equations.

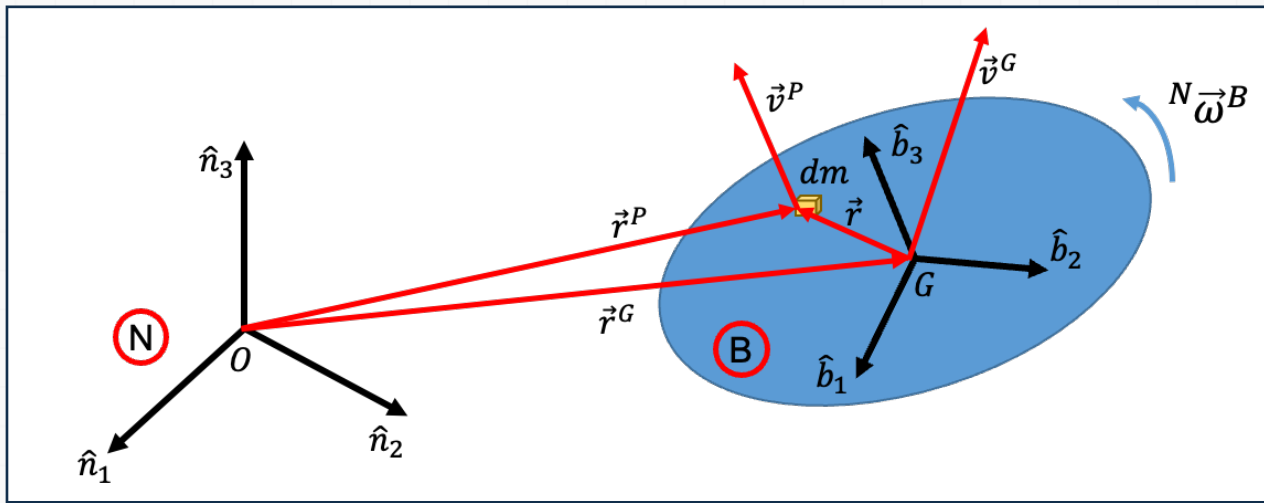
$$\vec{F} = m\vec{a} \quad \rightarrow \quad F = \frac{d\vec{L}}{dt}$$

where $\vec{L}$ is the linear momentum.

Linear Momentum for Translation

⚙️ Linear momentum

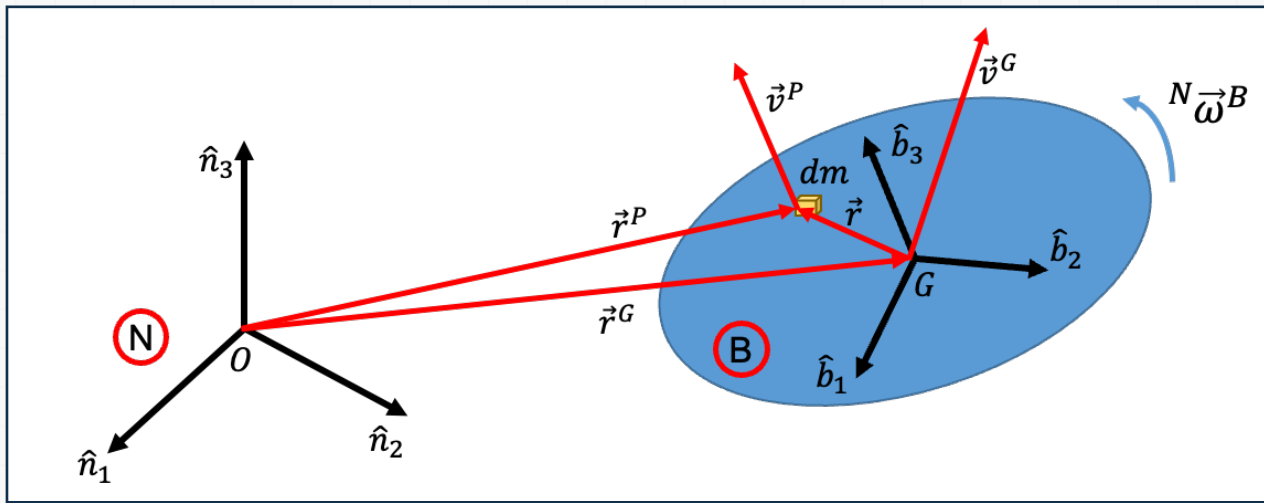
- Consider the following reference frame B with the angular velocity ${}^N\vec{\omega}^B$ and its center of mass, G .
 - Both the Newtonian reference frame N and the body B have their own coordinate systems $\hat{n}_1\hat{n}_2\hat{n}_3$ and $\hat{b}_1\hat{b}_2\hat{b}_3$, respectively.



Linear Momentum for Translation

⚙️ Linear momentum

- Also, consider a small mass element dm to examine a rigid body with random shapes.
- Note that we have the corresponding position vectors and velocity vectors.



Linear Momentum for Translation

⚙ Linear momentum of dm

$$dm \vec{v}^P$$

⚙ Total linear momentum of B

$$\begin{aligned} & \int_B dm \vec{v}^P \\ &= \int_B (\vec{v}^G + {}^N\vec{\omega}^B \times \vec{r}) dm = \int_B \vec{v}^G dm + \int_B {}^N\vec{\omega}^B \times \vec{r} dm \\ &= \int_B \vec{v}^G dm + {}^N\vec{\omega}^B \times \int_B \vec{r} dm = \int_B \vec{v}^G dm + 0 = m\vec{v}^G \end{aligned}$$

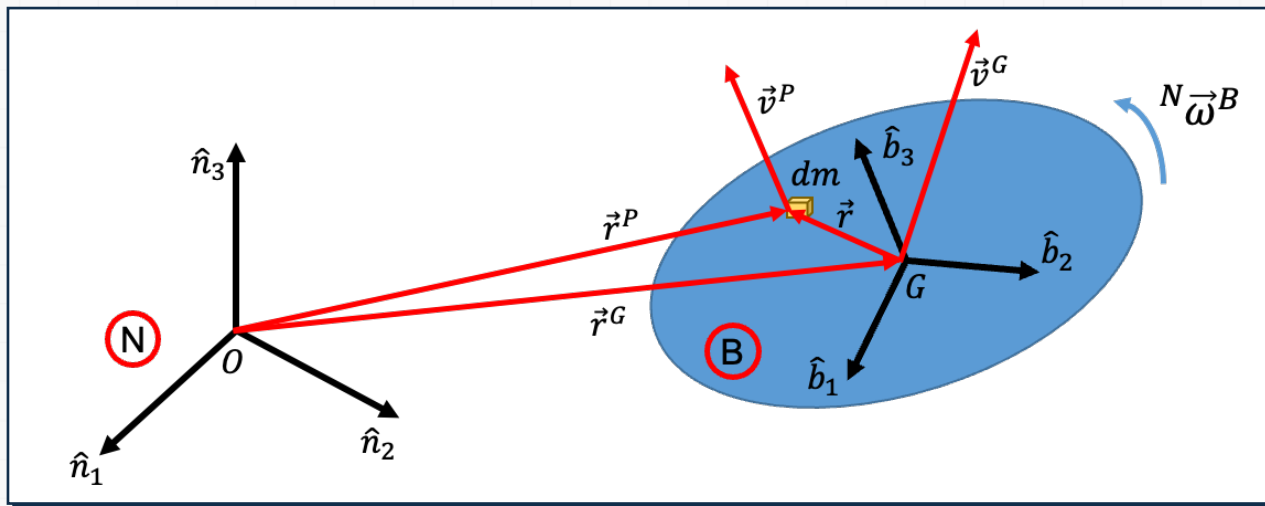
Linear Momentum for Translation

Linear momentum of a rigid body is the same as the linear momentum of its COM.

Angular Momentum for Rotation about G

⚙ Angular momentum

- Since a rigid body may also rotate, angular momentum should be considered as well.



Angular Momentum for Rotation about G

⚙ Angular momentum of dm about G

$$\vec{r} \times dm \vec{v}^P$$

⚙ Total angular momentum of B about G

$$\begin{aligned} \int_B \vec{r} \times dm \vec{v}^P &= \int_B \vec{r} \times \vec{v}^P dm = \int_B \vec{r} \times (\vec{v}^G + {}^N\vec{\omega}^B \times \vec{r}) dm \\ &= \int_B \vec{r} \times \vec{v}^G dm + \int_B \vec{r} \times {}^N\vec{\omega}^B \times \vec{r} dm = \left(\int_B \vec{r} dm \right) \times \vec{v}^G + \int_B \vec{r} \times {}^N\vec{\omega}^B \times \vec{r} dm \\ &= \int_B \vec{r} \times ({}^N\vec{\omega}^B \times \vec{r}) dm \end{aligned}$$

Angular Momentum for Rotation about G

⚙️ Introduction of a body coordinate system

- Not link the linear momentum, the last expression is not a convenient solution.
- To find more practical solution, let's introduce coordinates in terms of body.

$$\vec{r} = r_1 \hat{b}_1 + r_2 \hat{b}_2 + r_3 \hat{b}_3$$

$${}^N\vec{\omega}^B = \omega_1 \hat{b}_1 + \omega_2 \hat{b}_2 + \omega_3 \hat{b}_3$$

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Angular Momentum and Moment of Inertia

⚙ After introducing coordinates for $\vec{r}$ and ${}^N\vec{\omega}^B$ the total angular momentum of B can be computed.

- Total angular momentum of B about G

$$\begin{aligned}
 & \int_B \vec{r} \times ({}^N\vec{\omega}^B \times \vec{r}) dm \\
 & \int_B \begin{pmatrix} [(r_2^2 + r_3^2)\omega_1 - r_1r_2\omega_2 - r_1r_3\omega_3]\hat{b}_1 \\ + [-r_1r_2\omega_1 + (r_1^2 + r_3^2)\omega_2 - r_2r_3\omega_3]\hat{b}_2 \\ + [-r_1r_3\omega_1 - r_2r_3\omega_2 + (r_1^2 + r_2^2)\omega_3]\hat{b}_3 \end{pmatrix} dm = + \begin{pmatrix} \left(\int_B [(r_2^2 + r_3^2)\omega_1 - r_1r_2\omega_2 - r_1r_3\omega_3] dm \right) \hat{b}_1 \\ \left(\int_B [-r_1r_2\omega_1 + (r_1^2 + r_3^2)\omega_2 - r_2r_3\omega_3] dm \right) \hat{b}_2 \\ + \left(\int_B [-r_1r_3\omega_1 - r_2r_3\omega_2 + (r_1^2 + r_2^2)\omega_3] dm \right) \hat{b}_3 \end{pmatrix} \\
 & = H_1\hat{b}_1 + H_2\hat{b}_2 + H_3\hat{b}_3
 \end{aligned}$$

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Angular Momentum and Moment of Inertia

⚙ After introducing coordinates for $\vec{r}$ and ${}^N\vec{\omega}^B$ the total angular momentum of B can be computed.

- Total angular momentum of B about G

$$= \begin{pmatrix} H_1 \\ H_2 \\ H_3 \end{pmatrix}_B = \begin{pmatrix} \int_B (r_2^2 + r_3^2) dm & -\int_B r_1 r_2 dm & -\int_B r_1 r_3 dm \\ -\int_B r_2 r_1 dm & \int_B (r_1^2 + r_3^2) dm & -\int_B r_2 r_3 dm \\ -\int_B r_3 r_1 dm & -\int_B r_3 r_2 dm & \int_B (r_1^2 + r_2^2) dm \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = [I_G]_B [\omega]_B = I_G \omega$$

Abuse of notation

Notational convenience

Moment of Inertia Matrix

⚙️ **Total angular momentum of B about G is product of moment of inertia matrix and angular velocity.**

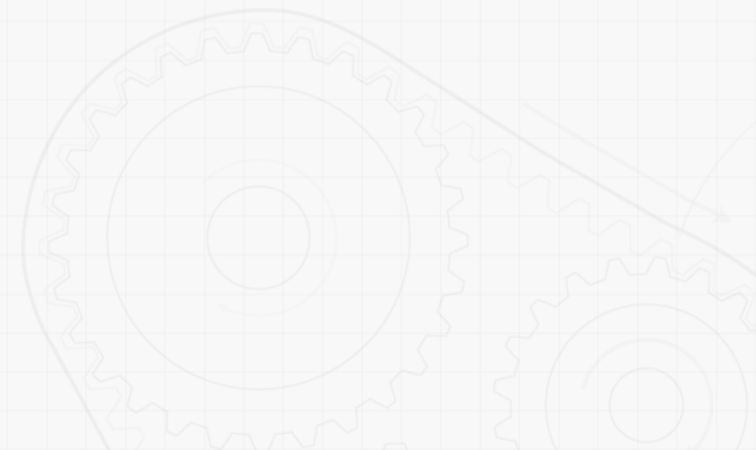
• Moment of inertia matrix

$$\begin{pmatrix} \int_B (r_2^2 + r_3^2) dm & -\int_B r_1 r_2 dm & -\int_B r_1 r_3 dm \\ -\int_B r_2 r_1 dm & \int_B (r_1^2 + r_3^2) dm & -\int_B r_2 r_3 dm \\ -\int_B r_3 r_1 dm & -\int_B r_3 r_2 dm & \int_B (r_1^2 + r_2^2) dm \end{pmatrix}_B$$

Moment of Inertia Matrix

⚙ Moment of inertia

- Moment of inertia is the resistance of the body to rotate.



Moment of Inertia Matrix

⚙ Feature of moment of inertia

- Moment of inertia matrix is **dependent on the geometric configuration of the rigid body**.
- Moment of inertia matrix is also **dependent on the choice of coordinate systems**.
 - If the chosen coordinate system is fixed to the body, then moment of inertia matrix is constant.
 - Otherwise, moment of inertia matrix changes as the body rotates.
- Moment of inertia matrix is **symmetric**.

Moment of Inertia Matrix

⚙️ Appropriate coordinate systems

- Appropriate coordinate system can drop the off-diagonal terms.
- If $\vec{r}$ is expressed with different coordinate systems,
 ${}^N\vec{\omega}^B$ should also be expressed in the same coordinate vectors.

Parallel Axis Theorem

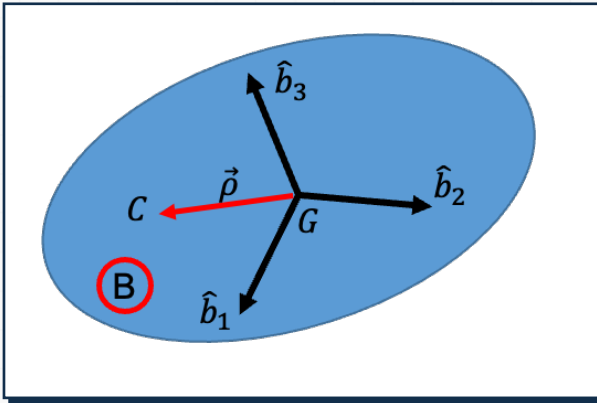
- ⚙️ If an angular momentum is computed about any arbitrary point other than G
 - then the moment of inertia should be redefined accordingly.

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Parallel Axis Theorem

⚙️ **Parallel axis theorem can ease the process.**

- Moment of inertia matrix about C



$$I_C = I_G + m[(\vec{\rho} \cdot \vec{\rho})I_3 - \vec{\rho} \otimes \vec{\rho}]$$

$$= I_G + m \begin{pmatrix} \rho_1^2 + \rho_2^2 + \rho_3^2 & -\rho_1\rho_2 & -\rho_1\rho_3 \\ -\rho_2\rho_1 & \rho_1^2 + \rho_2^2 + \rho_3^2 & -\rho_2\rho_3 \\ -\rho_3\rho_1 & -\rho_3\rho_2 & \rho_1^2 + \rho_2^2 + \rho_3^2 \end{pmatrix}$$

- Where $\vec{\rho} = \rho_1\hat{b}_1 + \rho_2\hat{b}_2 + \rho_3\hat{b}_3$, $\otimes$ is the outer product, I_3 is the 3-by-3 identity matrix.

Newton's Law of Motion for Translation

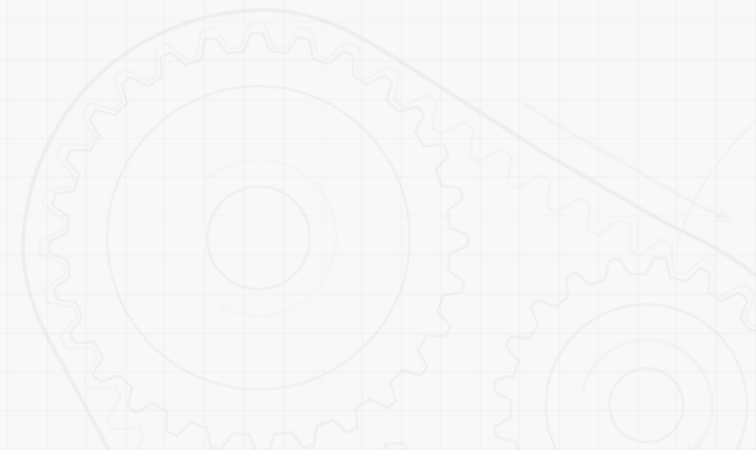
Newton's law of motion for particles can be extended for rigid bodies in 3D space.

$$\sum_i \vec{F}_i = {}^N \dot{\vec{L}}$$

$$\text{where } \vec{L} = m\vec{v}_G$$

Euler's Law of Motion for Rotation about COM

Euler's law of motion extends Newton's law of motion for particles to rigid body motion.



Euler's Law of Motion for Rotation about COM G

⚙️ For rotation, the following formulation is used.

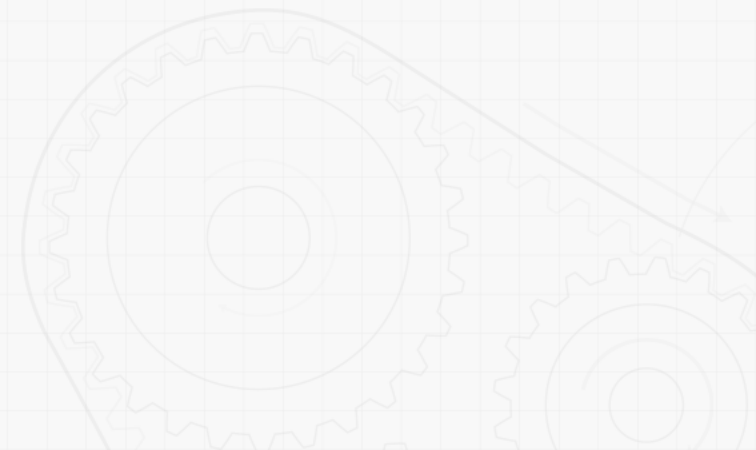
- If rotation is about the COM, G

$$\sum_i \vec{M}_{G,i} = {}^N \dot{\vec{H}}_G$$

where $\vec{H}_G = H_1 \hat{b}_1 + H_2 \hat{b}_2 + H_3 \hat{b}_3 = [I_G]_B [\omega]_B = I_G \omega$

↓
Abuse of notation

↓
Notational convenience



Euler's Law of Motion for Rotation about COM G

⚙️ For rotation, the following formulation is used.

- Since the differentiation should be done in N not B , use **vector differentiation formula**.

$${}^N \dot{\vec{H}}_G = {}^B \dot{\vec{H}}_G + {}^N \vec{\omega}^B \times \vec{H}_G$$

Euler's Law of Motion for Rotation about COM G

⚙️ For rotation, the following formulation is used.

• Therefore,

Abuse of notation
↓

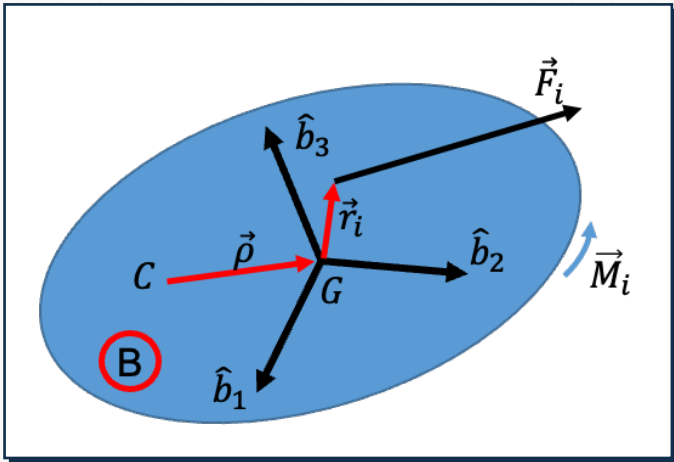
$$\sum_i M_{G,i} = {}^B\dot{\vec{H}}_G + {}^N\vec{\omega}^B \times \vec{H}_G = I_G \dot{\omega} + \omega \times I_G \omega = I_G \dot{\omega} + \hat{\omega} I_G \omega$$

where $\hat{\omega}$ is the skew symmetric matrix for ω .

$$\hat{\omega} = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}$$

Total Moment Considering External Forces

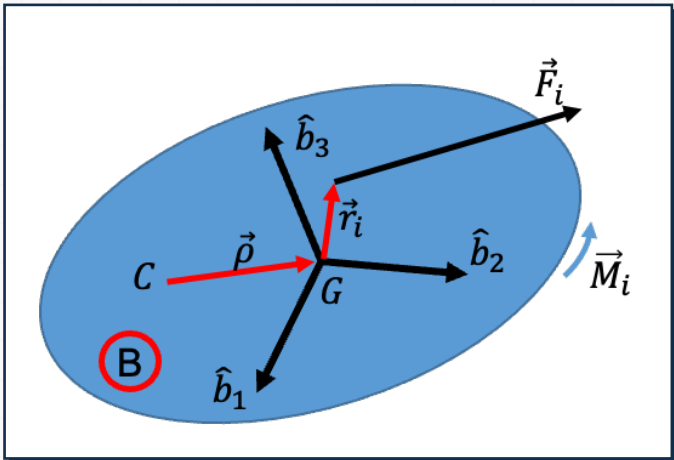
- ⚙️ **Note that external forces that do not pass through the point of rotation also contribute to the total moment.**
- Therefore, the total moment about COM G should be revised as follows.



$$\sum_i \vec{M}_G = \sum_i \vec{M}_i + \sum_i \vec{r}_i \times \vec{F}_i$$

Total Moment Considering External Forces

- ⚙️ **Note that external forces that do not pass through the point of rotation also contribute to the total moment.**
- If the rotation is about any arbitrary point C , the total moment about C should be as follows.



$$\sum_i \vec{M}_C = \sum_i \vec{M}_i + \sum_i (\vec{\rho}_i + \vec{r}_i) \times \vec{F}_i$$

Euler's Law of Motion for Rotation about C

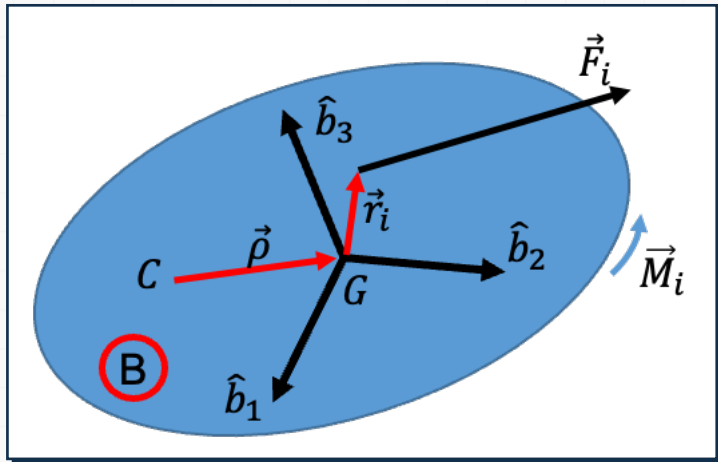
⚙️ If rotation is about any arbitrary point C

$$\begin{aligned}\sum_i M_{C,i} &= \sum_i \vec{M}_i + \sum_i (\vec{\rho}_i + \vec{r}_i) \times \vec{F}_i \\ &= \sum_i \vec{M}_i + \sum_i \vec{r}_i \times \vec{F}_i + \sum_i \vec{\rho}_i \times \vec{F}_i \\ &= \sum_i \vec{M}_G + \sum_i \vec{\rho}_i \times \vec{F}_i \\ &= \sum_i \vec{M}_G + \vec{\rho}_i \times \sum_i \vec{F}_i \\ &= \sum_i \vec{M}_G + \vec{\rho}_i \times m \vec{a}_G \\ &= \sum_i \vec{M}_G + \vec{\rho}_i \times \dot{\vec{L}}\end{aligned}$$

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Angular Momentum about Arbitrary Point C

⚙️ Note that the angular momentum about the COM G is computed as follows.

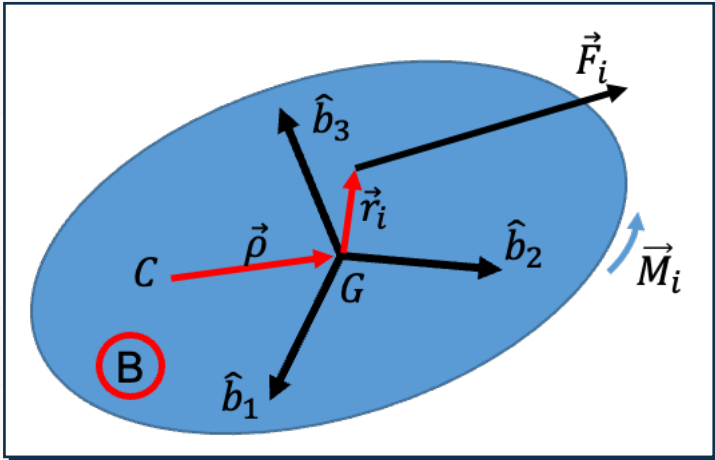


$$\vec{H}_G = I_G \omega$$

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Angular Momentum about Arbitrary Point C

⚙ The angular momentum about an arbitrary point C is computed as follows.

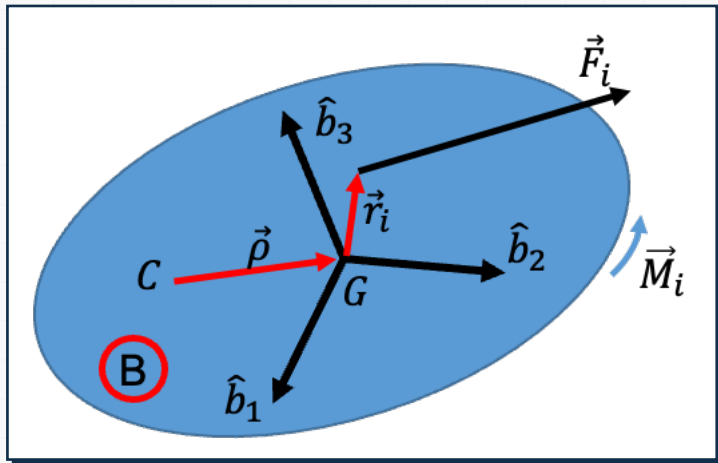


$$\begin{aligned}\vec{H}_C &= \int_B (\vec{\rho} + \vec{r}) \times dm \vec{v}^P \\ &= \dots \\ &= \vec{H}_G + \vec{\rho} \times \vec{L}\end{aligned}$$

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Angular Momentum about Arbitrary Point C

⚙ Note that



$$\vec{H}_C \neq I_C \omega$$

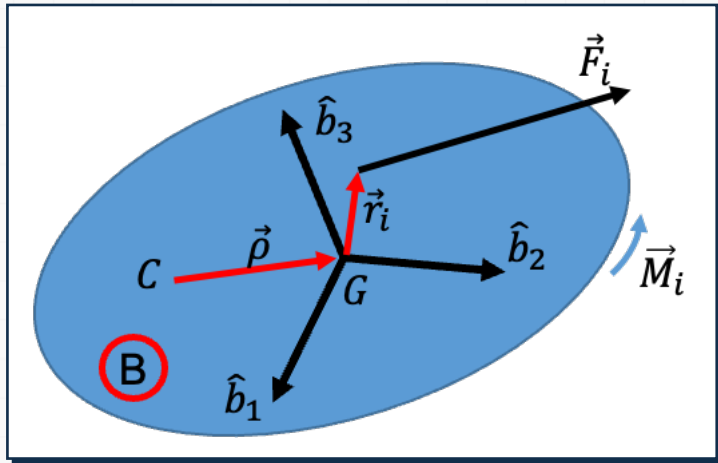
$$\begin{aligned} \vec{H}_C &= \vec{H}_G + \vec{\rho} \times \vec{L} = \vec{H}_G + \vec{\rho} \times m \vec{v}_G \\ &= \vec{H}_G + \vec{\rho} \times m (\vec{v}_C + {}^N\vec{\omega}^B \times \vec{\rho}) \\ &= I_G \omega + m \vec{\rho} \times {}^N\vec{\omega}^B \times \vec{\rho} + \vec{\rho} \times m \vec{v}_C \\ &= I_C \omega + \vec{\rho} \times m \vec{v}_C \end{aligned}$$

Only when C is a fixed point,

$$\vec{H}_C = I_C \omega$$

Euler's Law of Motion for Rotation about C (ctnd)

⚙️ If rotation is about any arbitrary point C



$$\begin{aligned}
 {}^N\dot{\vec{H}}_C &= \frac{d}{dt} ({}^N\vec{H}_G + m\vec{\rho} \times \vec{v}_G) \\
 &= \dots \\
 &= {}^N\dot{\vec{H}}_G + m\vec{\rho} \times \vec{a}_G + m\vec{\rho} \times \vec{v}_G \\
 &= \sum_i M_{C,i} + m\vec{v}_G \times \vec{v}_C
 \end{aligned}$$

• Only when C is a fixed point or the object is translating without rotation,

$$\sum_i M_{C,i} = {}^N\dot{\vec{H}}_C$$

Summary of Newton-Euler Equations

⚙ For translational motion

$$\sum_i \vec{F}_i = {}^N \dot{\vec{L}}, \quad \text{where } \vec{L} = m\vec{v}_G$$

⚙ For rotational motion about COM, G

$$\sum_i \vec{M}_{G,i} = {}^N \dot{\vec{H}}_G, \quad \text{where } \vec{H}_G = I_G \omega$$

Summary of Newton-Euler Equations

⚙️ For rotational motion about arbitrary point, C

$${}^N\dot{\vec{H}}_C = \sum_i \vec{M}_{C,i} + m\vec{v}_G \times \vec{v}_C, \quad \text{where } \vec{H}_C = I_C\omega + \vec{\rho} \times m\vec{v}_C$$

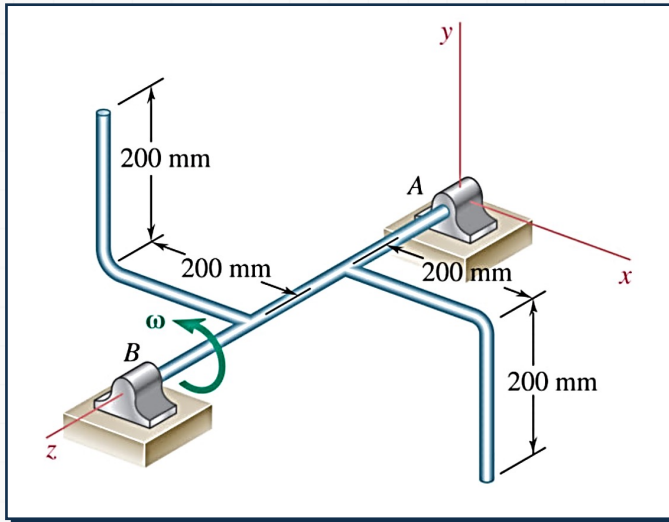
• If C is a fixed point,

$${}^N\dot{\vec{H}}_C = \sum_i \vec{M}_{C,i}, \quad \text{where } \vec{H}_C = I_C\omega$$

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Sample Problem 1

⚙️ Two L-shaped arms, each weighing 5kg, are welded at the third points of the 600mm shaft AB .



• Determine

- Knowing that shaft AB rotates at the constant rate $\omega = 360rpm$, determine the angular momentum of the body about A and the angle it forms with AB .

Sample Problem 1

⚙️ **Two L-shaped arms, each weighing 5kg, are welded at the third points of the 600mm shaft AB .**

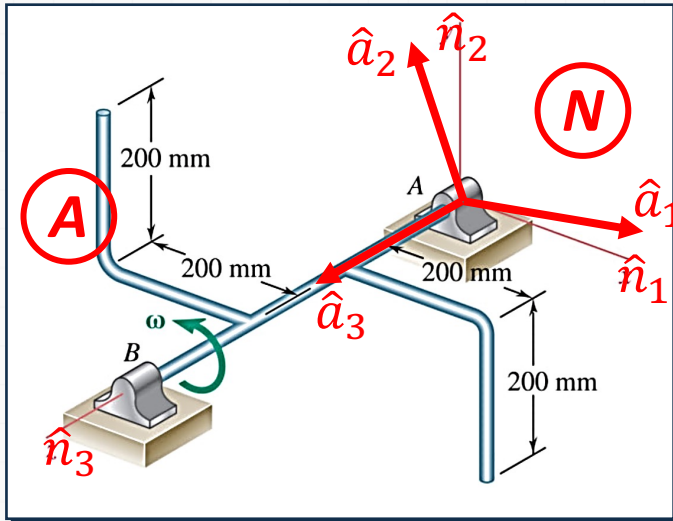
• Strategy

- Define reference frames and appropriate coordinate systems.
- The part rotates about the axis AB . Determine angular velocity of the arm.
- Determine the mass moment of inertia matrix for the part.
- Compute the angular momentum of the disk using the moments inertia.

12 Kinetics of Rigid Bodies

Sample Problem 1

⚙️ Define reference frames and coordinate systems.



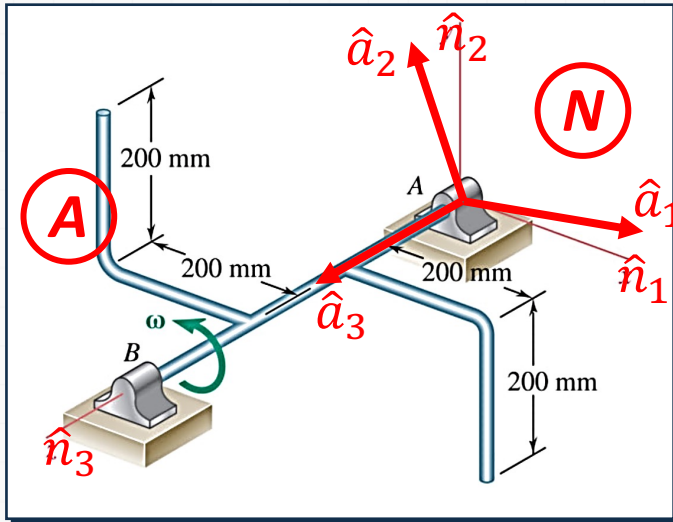
Inertial reference frame: N
with $\hat{n}_1 \hat{n}_2 \hat{n}_3$

Reference frame of arm: A
with $\hat{a}_1 \hat{a}_2 \hat{a}_3$

12 Kinetics of Rigid Bodies

Sample Problem 1

⚙ Determine angular velocity of the arm.



$$360 \text{ rpm} = \frac{360(2\pi)}{60} = 12\pi \text{ rad/s}$$

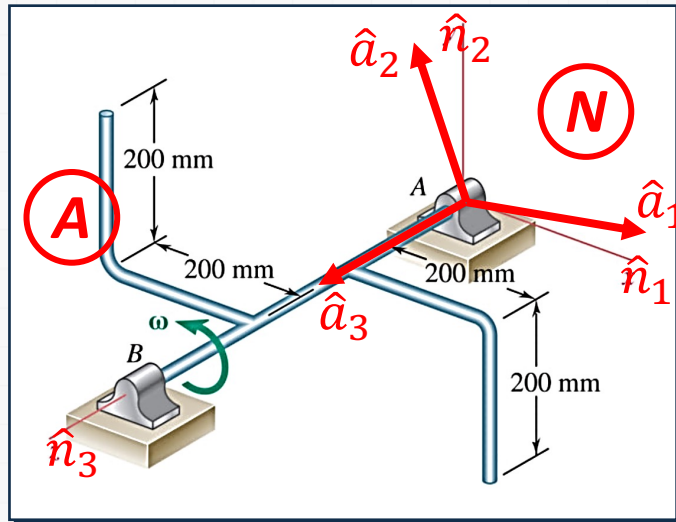
Angular velocity of the arm

$${}^N\vec{\omega}^A = 12\pi \hat{a}_3$$

12 Kinetics of Rigid Bodies

Sample Problem 1

⚙ Determine the moment of inertia matrix



$$I_G = \begin{pmatrix} \int_B (r_2^2 + r_3^2) dm & -\int_B r_1 r_2 dm & -\int_B r_1 r_3 dm \\ -\int_B r_2 r_1 dm & \int_B (r_1^2 + r_3^2) dm & -\int_B r_2 r_3 dm \\ -\int_B r_3 r_1 dm & -\int_B r_3 r_2 dm & \int_B (r_1^2 + r_2^2) dm \end{pmatrix}_B$$

Let's first compute $\vec{H}_G$. Then, $\vec{H}_A = \vec{H}_G + \vec{AG} \times m\vec{v}_G = \vec{H}_G$

$$\vec{H}_G = I_G \omega = \begin{pmatrix} X & X & -\int_B r_1 r_3 dm \\ X & X & -\int_B r_2 r_3 dm \\ X & X & \int_B (r_1^2 + r_2^2) dm \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 12\pi \end{pmatrix}$$

$$\int_B (r_1^2 + r_2^2) dm = \frac{5}{12} (0.4)^2 + 0.2^2 \times \frac{5}{2} \times 2 + \frac{5}{12} (0.4)^2 = \frac{1}{3}$$

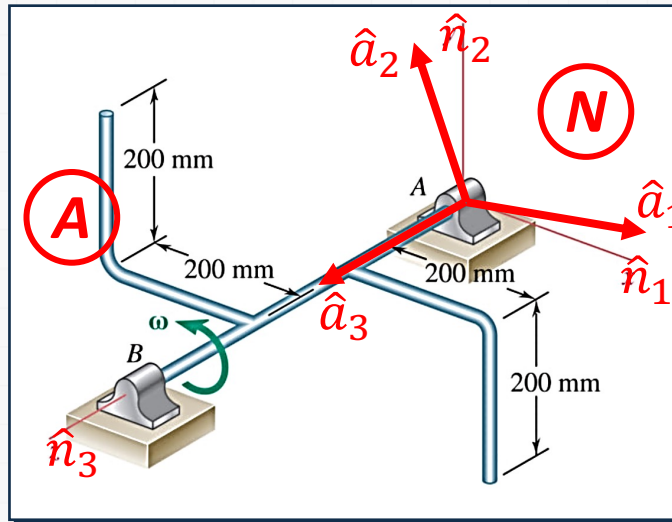
$$-\int_B r_1 r_3 dm = -\int_{-0.2}^0 0.1 r_1 dr \frac{2.5}{0.2} - \int_0^{0.2} -0.1 r_1 dr \frac{2.5}{0.2} - 0.1(-0.2)2.5 + 0.1(0.2)2.5 = 0.15$$

$$-\int_B r_2 r_3 dm = -\int_{-0.2}^0 0.1 r_1 dr \frac{2.5}{0.2} - \int_0^{0.2} -0.1 r_1 dr \frac{2.5}{0.2} - 0.1(-0.2)2.5 + 0.1(0.2)2.5 = -0.05$$

12 Kinetics of Rigid Bodies

Sample Problem 1

⚙ Determine the moment of inertia matrix



Let's first compute $\vec{H}_G$. Then, $\vec{H}_A = \vec{H}_G + \vec{AG} \times m\vec{V}_G = \vec{H}_G$

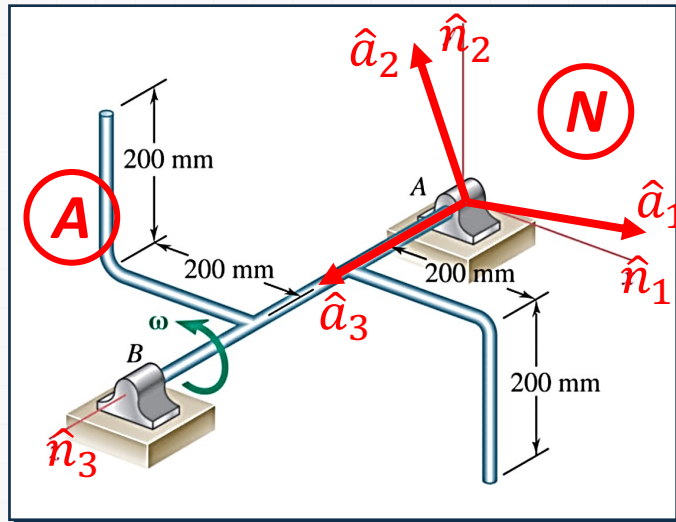
$$\vec{H}_G = I_G \omega = \begin{pmatrix} X & X & 0.15 \\ X & X & -0.05 \\ X & X & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 12\pi \end{pmatrix} = \begin{pmatrix} 5.6549 \\ 1.8849 \\ 12.5664 \end{pmatrix}$$

$$\rightarrow \vec{H}_A = 5.6549\hat{a}_3 - 1.8849\hat{a}_3 + 12.5664\hat{a}_3$$

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Sample Problem 1

⚙ Determine the angle between $\overrightarrow{AB}$ and $\vec{H}_A$

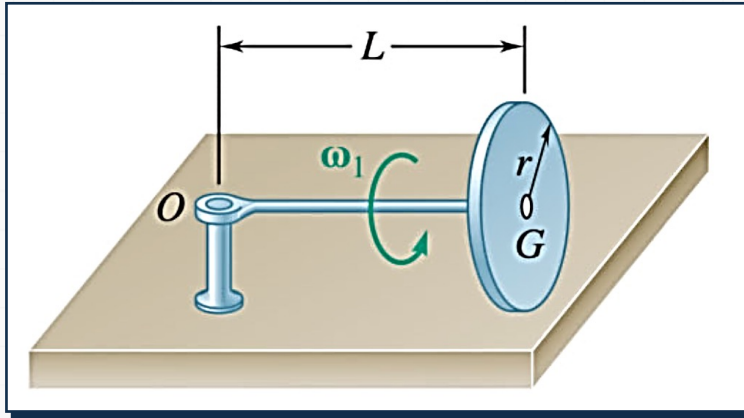


$$\cos \theta = \frac{\vec{H}_A \cdot \hat{a}_3}{\|\vec{H}_A\|} = \frac{12.5664}{13.9085}$$

$$\rightarrow \theta = 25.38^\circ$$

Sample Problem 2

⚙️ A homogeneous disk of mass m is mounted on an axle OG of negligible mass. The disk rotates counterclockwise at the rate ω_1 about OG .



• Determine

- a) the angular velocity of the disk
- b) its angular momentum about O

Sample Problem 2

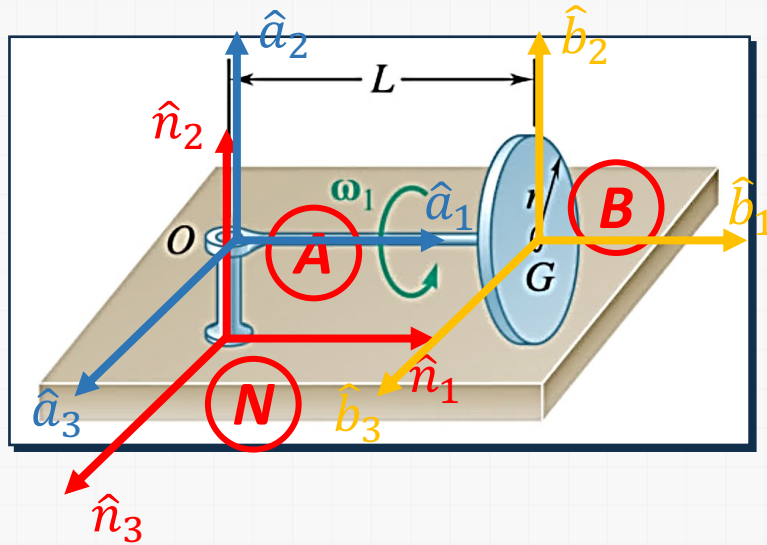
⚙️ **A homogeneous disk of mass m is mounted on an axle OG of negligible mass. The disk rotates counterclockwise at the rate ω_1 about OC .**

• Strategy

- Define reference frames and appropriate coordinate systems.
- The disk rotates about the vertical axis through O as well as about OG .
- The disk rolls, and thus the contact point with the ground has zero velocity.
- Combine the rotation components for the angular velocity of the disk.
- Compute the angular momentum of the disk using principle axes of inertia and noting that O is a fixed point.

Sample Problem 2

⚙ Define reference frames and coordinate systems.



Inertial reference frame: N
with $\hat{n}_1 \hat{n}_2 \hat{n}_3$

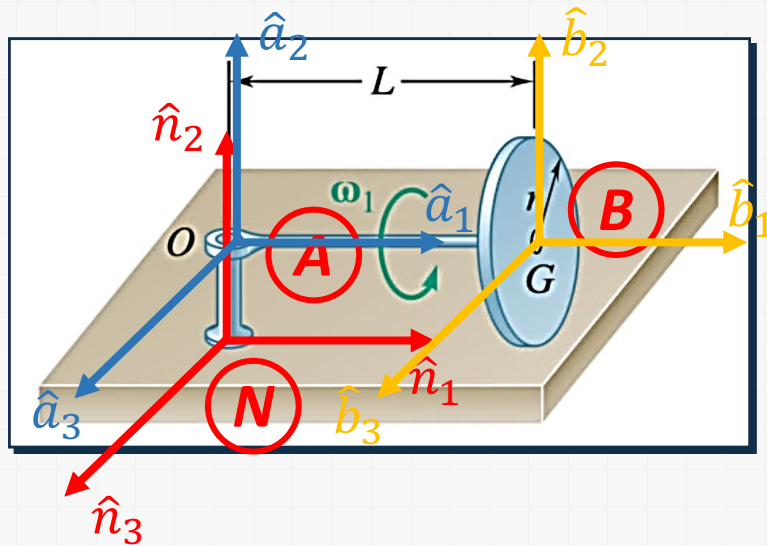
Reference frame of arm: A
with $\hat{a}_1 \hat{a}_2 \hat{a}_3$

Reference frame of arm: B
with $\hat{b}_1 \hat{b}_2 \hat{b}_3$

12 Kinetics of Rigid Bodies

Sample Problem 2

⚙️ Determine angular velocities.



Angular velocity of B with respect to A

$${}^A\vec{\omega}^B = \omega_1 \hat{a}_1$$

Assume that the axle rotates with angular velocity of $\omega_2 \hat{n}_2$.

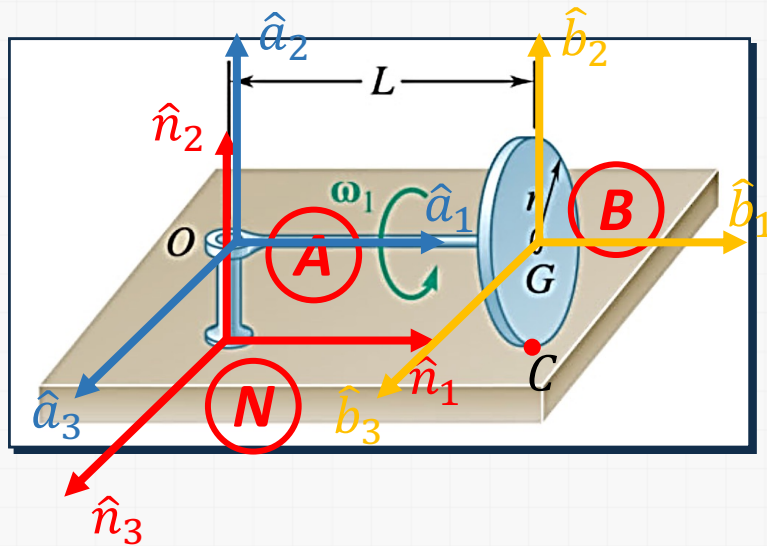
$${}^N\vec{\omega}^A = \omega_2 \hat{n}_2$$

Then, the angular velocity of B with respect to N is

$${}^N\vec{\omega}^B = \omega_2 \hat{n}_2 + \omega_1 \hat{a}_1$$

Sample Problem 2

⚙ Determine angular velocities.



The contact point C of the disk at the ground has zero velocity. Using two-point relation,

$$\begin{aligned}\vec{v}^C &= \vec{v}^O + {}^N\vec{\omega}^B \times \overrightarrow{OC} \\ &= 0 + (\omega_2 \hat{n}_2 + \omega_1 \hat{a}_1) \times (L \hat{b}_1 - r \hat{b}_2)\end{aligned}$$

Note that at this moment, $\hat{a}_1 = \hat{b}_1, \hat{n}_2 = \hat{b}_2$

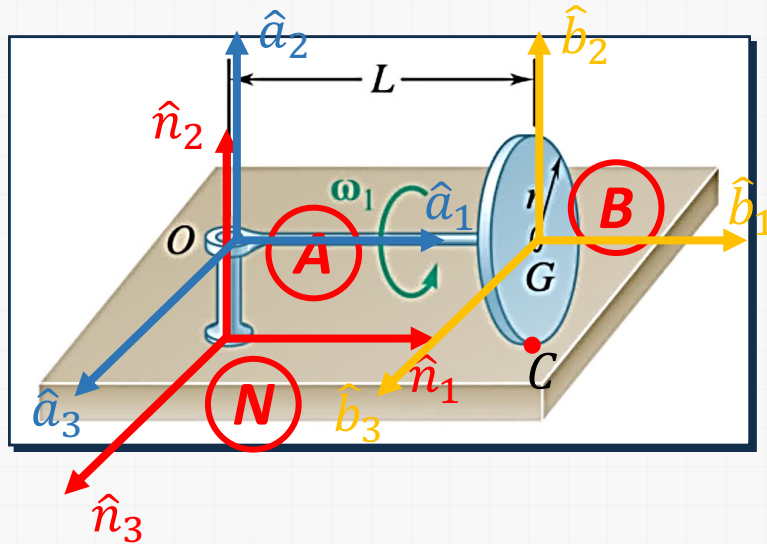
$$\begin{aligned}\vec{v}^C &= (\omega_2 \hat{b}_2 + \omega_1 \hat{b}_1) \times (L \hat{b}_1 - r \hat{b}_2) \\ &= -(L\omega_2 + r\omega_1) \hat{b}_3 \\ &= 0\end{aligned}$$

Therefore, $\omega_2 = -\frac{r\omega_1}{L}$

$${}^N\vec{\omega}^B = \omega_1 \hat{a}_1 + -\frac{r\omega_1}{L} \hat{n}_2$$

Sample Problem 2

⚙ Determine the moment of inertia of the disk



The moment of inertia matrix of the disk is

$$I_G = \begin{pmatrix} \frac{1}{2}mr^2 & 0 & 0 \\ 0 & \frac{1}{4}mr^2 & 0 \\ 0 & 0 & \frac{1}{4}mr^2 \end{pmatrix}$$

Note that at this moment, $\hat{a}_1 = \hat{b}_1, \hat{n}_2 = \hat{b}_2$

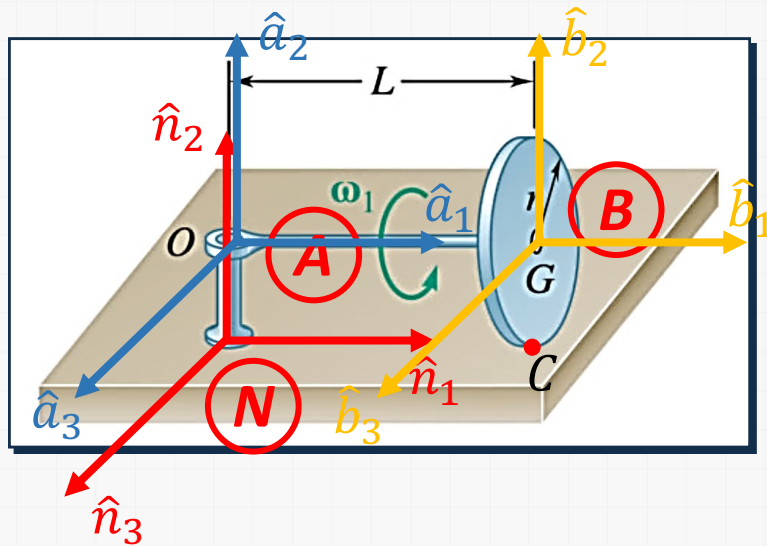
$${}^N\vec{\omega}^B = \omega_1 \hat{b}_1 + -\frac{r\omega_1}{L} \hat{b}_2$$

Therefore, the angular momentum is

$${}^N\vec{H}_G^B = \frac{1}{2}mr^2\omega_1\hat{b}_1 - \frac{1}{4}m\frac{r^3}{L}\omega_1\hat{b}_2$$

Sample Problem 2

⚙ Determine the angular momentum of the disk about O



The angular momentum of the disk about O can be computed as follows.

$${}^N\vec{H}_O^B = {}^N\vec{H}_G^B + \vec{OG} \times m\vec{v}^G$$

Therefore, the angular momentum is

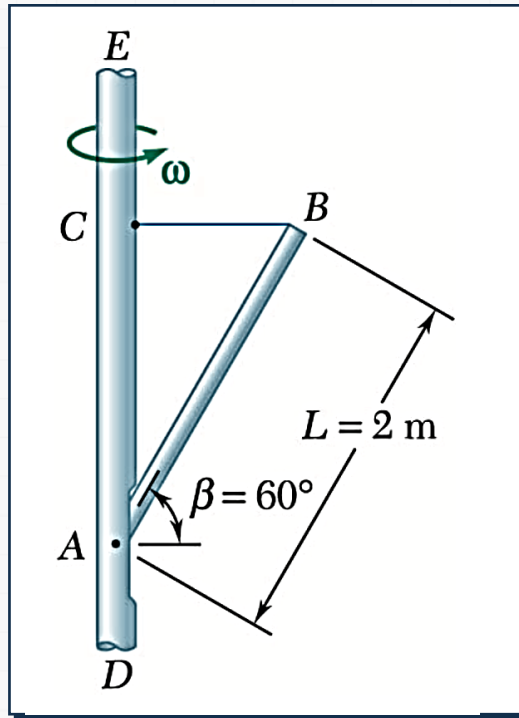
$$\begin{aligned}\vec{v}^G &= \left(\omega_1 \hat{b}_1 + -\frac{r\omega_1}{L} \hat{b}_2 \right) \times L \hat{b}_1 \\ &= r\omega_1 \hat{b}_3\end{aligned}$$

Putting together,

$$\begin{aligned}{}^N\vec{H}_O^B &= \frac{1}{2}mr^2\omega_1\hat{b}_1 - \frac{1}{4}m\frac{r^3}{L}\omega_1\hat{b}_2 + L\hat{b}_1 \times mr\omega_1\hat{b}_3 \\ &= \frac{1}{2}mr^2\omega_1\hat{b}_1 - \frac{mr\omega_1}{L}\left(\frac{1}{4}r^2 + L^2\right)\hat{b}_2\end{aligned}$$

Sample Problem 3

⚙️ Rod AB with mass $m=20\text{kg}$ is pinned at A to a vertical axle which rotates with constant angular velocity $\omega = 15 \text{ rad/s}$.



• Determine

- The rod position is maintained by a horizontal wire BC. Determine the tension in the wire and the reaction at A.

Sample Problem 3

⚙️ **Rod AB with mass $m=20\text{kg}$ is pinned at A to a vertical axle which rotates with constant angular velocity $\omega = 15 \text{ rad/s}$.**

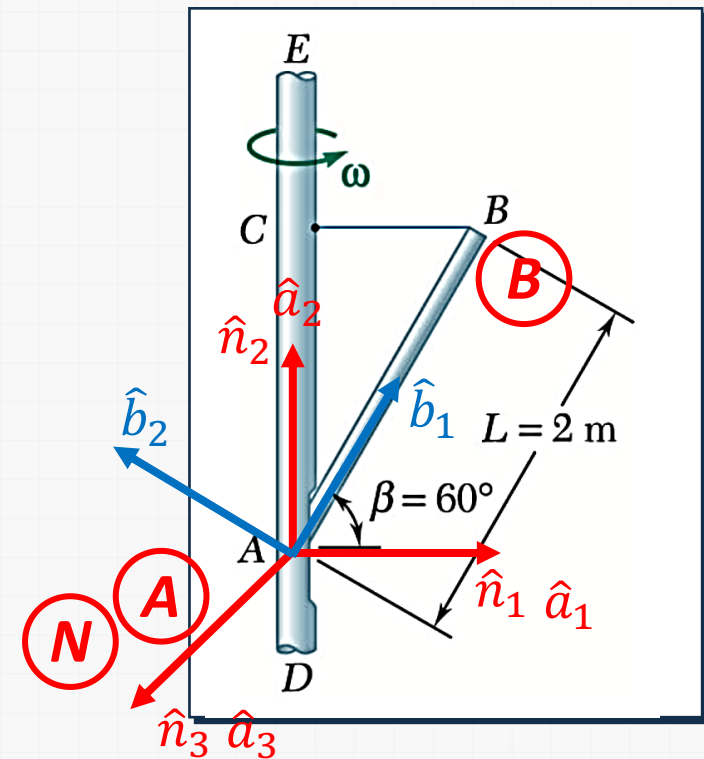
• Strategy

- Evaluate the system of effective forces by reducing them to a vector $m\vec{a}$ attached at G and couple $\vec{H}_G$.
- Expressing that the system of external forces is equivalent to the system of effective forces, write vector expressions for the sum of moments about A and the summation of forces.
- Solve for the wire tension and the reactions at A .

12 Kinetics of Rigid Bodies

Sample Problem 3

⚙ Define reference frames and coordinate systems.



Inertial reference frame: N
with $\hat{n}_1\hat{n}_2\hat{n}_3$

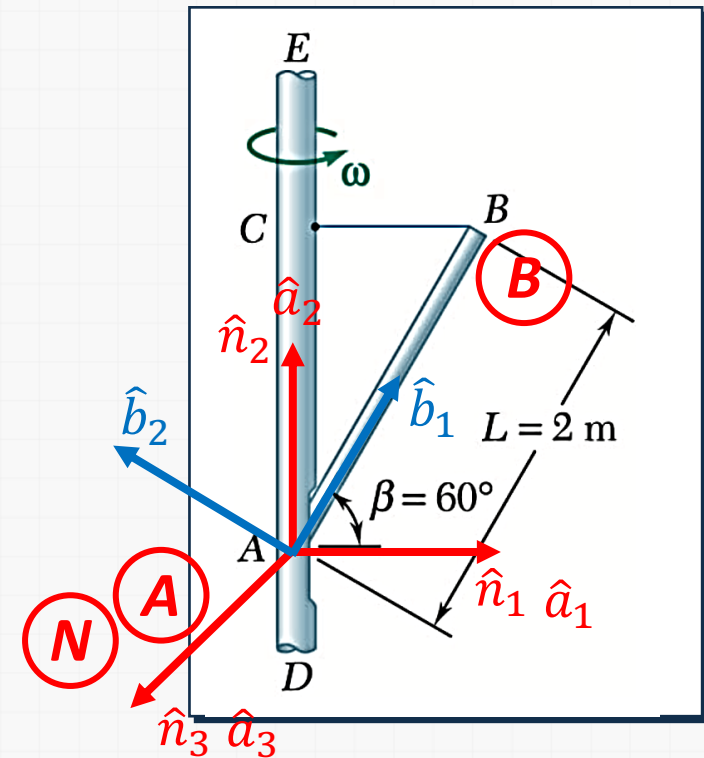
Reference frame of axle: A
with $\hat{a}_1\hat{a}_2\hat{a}_3$

Reference frame of rod: B
with $\hat{b}_1\hat{b}_2\hat{b}_3$

12 Kinetics of Rigid Bodies

Sample Problem 3

⚙️ Determine angular velocities.



Angular velocity of A with respect to N

$${}^N\vec{\omega}^A = \omega \hat{n}_2$$

Angular velocity of B with respect to A

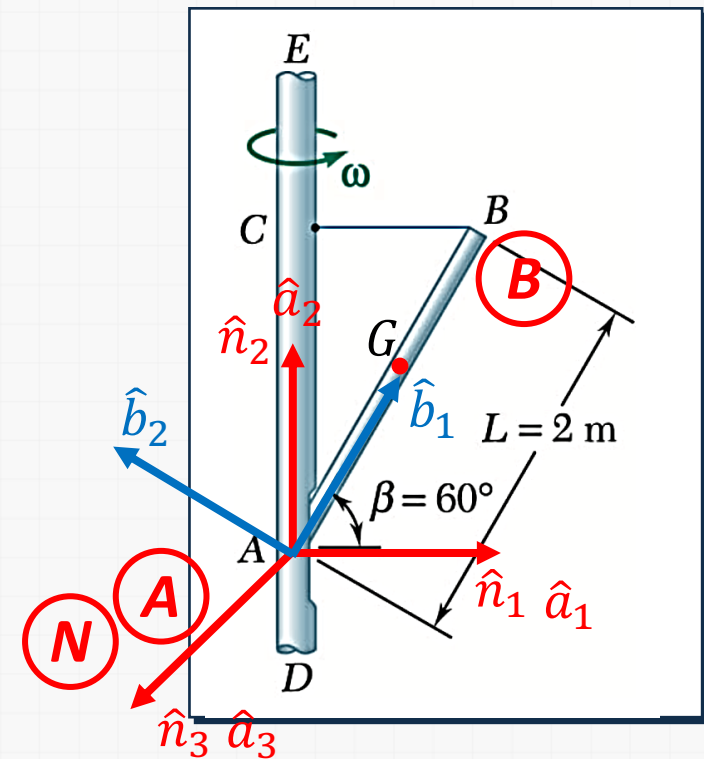
$${}^A\vec{\omega}^B = \dot{\beta} \hat{a}_3$$

Then, the angular velocity of B with respect to N is

$${}^N\vec{\omega}^B = \omega \hat{n}_2 + \dot{\beta} \hat{a}_3$$

Sample Problem 3

⚙ Determine the moment of inertia of the rod about COM G.



Moment of inertia of the rod about COM G

$$I_G = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{12}mL^2 & 0 \\ 0 & 0 & \frac{1}{12}mL^2 \end{pmatrix}$$

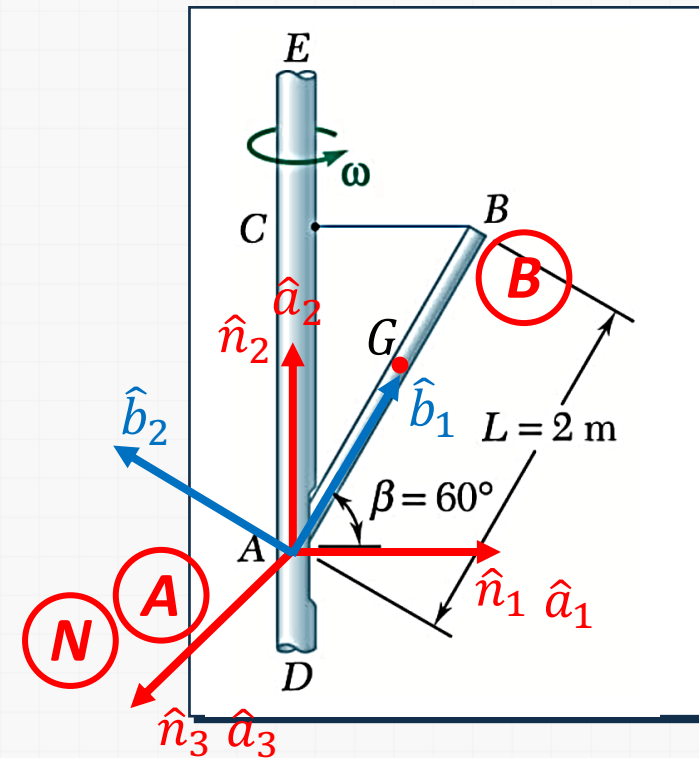
$$\begin{aligned} {}^N\vec{\omega}^B &= \omega \hat{n}_2 + \dot{\beta} \hat{b}_3 = \omega \hat{n}_2 \\ &= \omega \sin \beta \hat{b}_1 + \omega \cos \beta \hat{b}_2 \end{aligned}$$

Then, the angular momentum of the rod about COM G is

$${}^N\vec{H}_G^B = I_G {}^N\vec{\omega}^B = \frac{1}{12}mL^2 \omega \cos \beta \hat{b}_2$$

Sample Problem 3

⚙️ Determine the rate of change of the angular momentum.



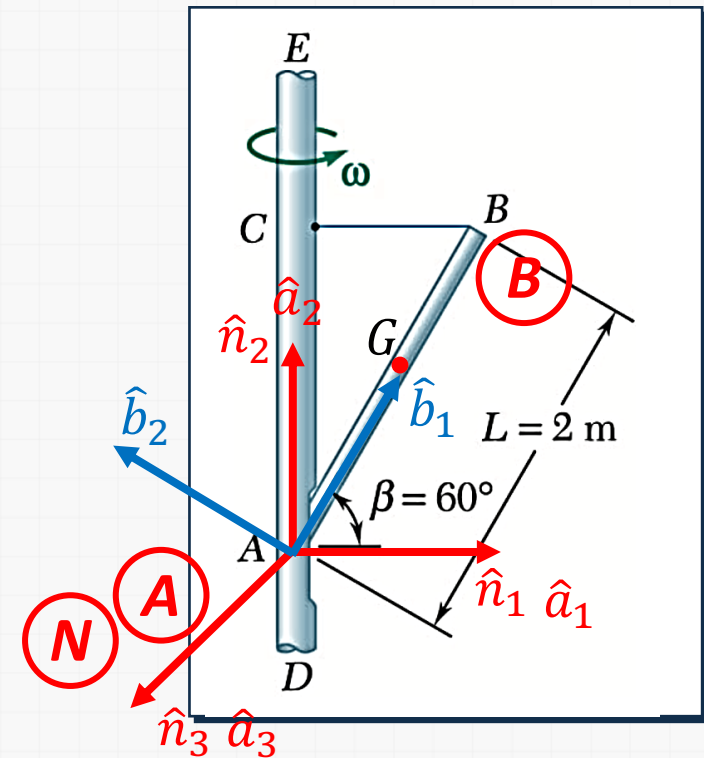
The angular momentum must be differentiated in Newtonian reference frame. Therefore, apply the vector differentiation formula.

$$\begin{aligned}
 {}^N\dot{\vec{H}}_G^B &= {}^B\dot{\vec{H}}_G^B + {}^N\vec{\omega}^B \times {}^N\vec{H}_G^B \\
 &= 0 + (\omega \sin \beta \hat{b}_1 + \omega \cos \beta \hat{b}_2) \times \frac{1}{12} mL^2 \omega \cos \beta \hat{b}_2 \\
 &= \frac{1}{12} mL^2 \omega^2 \sin \beta \cos \beta \hat{b}_3
 \end{aligned}$$

12 Kinetics of Rigid Bodies

Sample Problem 3

⚙ Determine the acceleration of the COM G



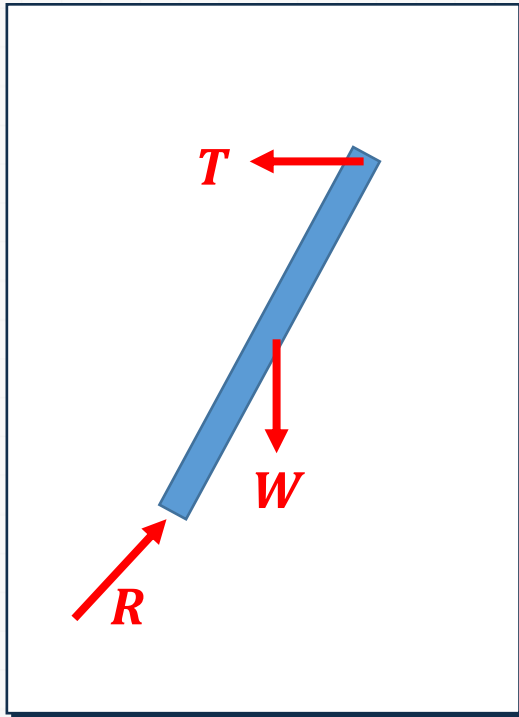
Using two-point relation, the acceleration of the rod at the COM G can be easily computed.

$$\begin{aligned}
 {}^N\vec{a}^G &= {}^N\vec{a}^A + {}^N\vec{\alpha}^B \times \overrightarrow{AG} + {}^N\vec{\omega}^B \times ({}^N\vec{\omega}^B \times \overrightarrow{AG}) \\
 &= 0 + 0 + (\omega \sin \beta \hat{b}_1 + \omega \cos \beta \hat{b}_2) \\
 &\quad \times \left((\omega \sin \beta \hat{b}_1 + \omega \cos \beta \hat{b}_2) \times \frac{1}{2} L \hat{b}_1 \right) \\
 &= -\frac{1}{2} \omega^2 \cos \beta L (\cos \beta \hat{b}_1 - \sin \beta \hat{b}_2) \\
 &= -\frac{1}{2} \omega^2 \cos \beta L \hat{n}_1
 \end{aligned}$$

12 Kinetics of Rigid Bodies

Sample Problem 3

⚙️ Apply Newton's and Euler's Laws



Newton's law for translation

$$\begin{aligned}\sum \vec{F} &= R_1 \hat{n}_1 + R_2 \hat{n}_2 + R_3 \hat{n}_3 - mg \hat{n}_2 - T \hat{n}_1 \\ &= (R_1 - T) \hat{n}_1 + (R_2 - mg) \hat{n}_2 + R_3 \hat{n}_3 \\ &= m {}^N \vec{a}^G = -\frac{1}{2} m \omega^2 \cos \beta L \hat{n}_1\end{aligned}$$

Euler's law for rotation

$$\begin{aligned}\sum \vec{M}_G &= \vec{GA} \times \vec{R} + \vec{GB} \times \vec{T} \\ &= -\frac{L}{2} \hat{b}_1 \times (R_1 \hat{n}_1 + R_2 \hat{n}_2 + R_3 \hat{n}_3) + \frac{L}{2} \hat{b}_1 \times (-T \hat{n}_1) \\ &= {}^N \dot{\vec{H}}_G^B = \frac{1}{12} m L^2 \omega^2 \sin \beta \cos \beta \hat{b}_3\end{aligned}$$

Solving together, we have

$$\vec{T} = -1557 \hat{n}_1, \vec{R} = -697 \hat{n}_1 + 196.2 \hat{n}_2 \text{ (N)}$$

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, McGraw Hill, Chapter 18
- 유홍희, "동역학", 1판, 문운당, 8장